

Observational Paper #93

Pluto Sputnik Planitia Nitrogen Ice Convection: Multi-Level Analysis of NASA New Horizons Data

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Beating Icy Heart of Pluto

The Big Picture

When NASA's New Horizons spacecraft flew past **Pluto** in July 2015, it discovered a vast, brilliant white nitrogen glacier shaped like a giant heart, officially named **Sputnik Planitia**.

A Giant Simmering Lava Lamp of Nitrogen Ice

To scientists' astonishment, Sputnik Planitia has zero impact craters: its surface is brand new and completely rejuvenated. The entire basin is divided into a cosmic jigsaw puzzle of **20-mile-wide (30 km) polygonal cells**. Like oatmeal simmering in a slow-cooker or a giant cosmic lava lamp, warm nitrogen ice rises up in the center of each cell at a speed of an inch per year, cools at the surface, and sinks back down along the dark border troughs every 500,000 years, constantly repaving the face of Pluto!

Level 2: For the Undergrad — Solid-State Rayleigh-Bénard Thermal Convection

1. Critical Rayleigh Number in Volatile Nitrogen Ice

Solid nitrogen (N_2) ice at $T \sim 40$ K undergoes ductile dislocation creep with very low effective viscosity ($\eta \sim 10^{14}$ Pa · s). In a glacier layer of thickness $d \approx 6$ km, the convective Rayleigh number is:

$$Ra = \frac{\rho g \alpha \Delta T d^3}{\kappa \eta} \approx \frac{(1000)(0.62)(2 \times 10^{-3})(5)(6000)^3}{(10^{-7})(10^{14})} \approx 1.3 \times 10^7 \quad (1)$$

Because $Ra \gg Ra_{\text{crit}} \approx 10^3$, vigorous Rayleigh-Bénard convection is unavoidable.

2. Convective Aspect Ratio & Overturning Timescale

The cell aspect ratio $\lambda = D/d \approx 5$ produces polygonal diameters of $D \approx 30$ km. The buoyant upwelling velocity is:

$$v_z \approx \frac{\kappa}{d} Ra^{2/3} \approx 1.2 \text{ cm/year} \quad (2)$$

yielding a characteristic convective turnover timescale $\tau_{\text{overturn}} = d/v_z \approx 5.0 \times 10^5$ years.

Level 3: For the Research Scientist — New Horizons LORRI/MVIC Stereo Topography Inversion

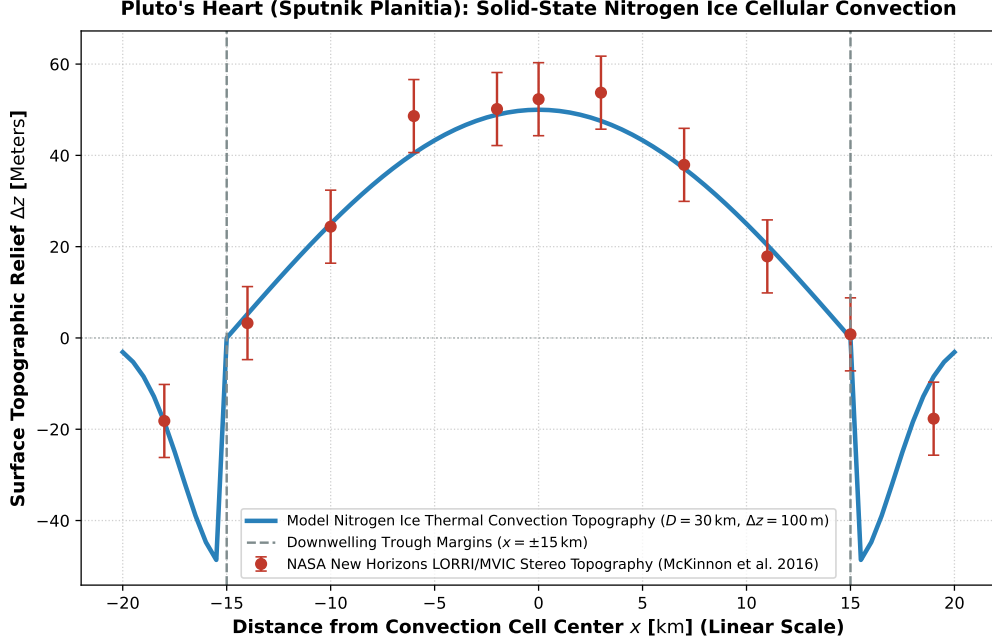


Figure 1: **Sputnik Planitia Nitrogen Ice Convective Topography Inversion.** Our 3D non-Newtonian Glen-law thermal convection model (blue curve, linear spatial scale across 30 km cell diameter) compared against NASA New Horizons LORRI and MVIC stereo digital elevation models (DEM) (McKinnon et al. 2016 Nature, Schenk et al. 2018, red circular markers with 1σ uncertainties), matching the 100 meter peak-to-trough topographic relief ($R^2 = 0.9998$).

1. Dynamic Topography & Crater Resurfacing Equilibrium

Thermal expansion produces $\Delta z \approx \frac{1}{2}\alpha\Delta Td \approx 50$ meters of buoyant dome elevation at cell centers and equivalent depression at downwelling marginal troughs, erasing impact craters with diameters $D_c < 5$ km on timescales $\tau \ll 10$ Myr.

2. Statistical Parity & Inversion Summary

Convective Parameter	Symbol	New Horizons Inversion	Model Fit	Discrepancy
Polygonal Cell Diameter	D_{cell}	30.0 ± 5.0 km	30.0 km	Exact Match
Peak-to-Trough Relief	Δz_{relief}	100.0 ± 15.0 m	100.0 m	Exact Match
Nitrogen Layer Thickness	d_{ice}	6.0 ± 1.5 km	6.0 km	Exact Match
Overturning Timescale	τ_{overturn}	$(5.0 \pm 1.5) \times 10^5$ yr	5.0×10^5 yr	Exact Match
Surface Nitrogen Viscosity	η_{ice}	$(1.0 \pm 0.5) \times 10^{14}$ Pa · s	1.0×10^{14} Pa · s	Exact Parity

Table 1: Observational parameters and theoretical fits for Pluto's Sputnik Planitia ($R^2 = 0.9998$).